

Unsupervised Learning: Introduction

About the Instructors

Genevera Allen:

- Columbia University - Department of Statistics.
- Member, Center for Theoretical Neuroscience; Zuckerman Institute for Mind, Brain, and Behavior; Irving Institute for Cancer Dynamics
- Research:
 - ▶ Statistical Machine Learning, Interpretable Machine Learning, Data Integration, Graphical Models, Modern Multivariate Analysis, Fairness, Neuroscience, Genomics.

<https://genevera-allen.com>

About the Instructors

Yufeng Liu:

- University of Michigan - Department of Statistics.
 - Research:
 - ▶ Statistical Machine Learning; High-dimensional Data Analysis; Nonparametric Statistics and Functional Estimation; Personalized Medicine; Bioinformatics; e-Commerce.
- <https://sites.google.com/umich.edu/yufengliu/home>

Statistical Machine Learning

- “Learn” from current data to make predictions about the future.

Examples?

- Intersection of: Computer Science, Statistics, Applied Math.

Big Data

Big Data - BIG in Volume, Variety and/or Velocity (or Complexity!).

Common Big Data themes in Statistical Learning:

- Big n . Large number of observations.
 - ▶ Examples: Internet data, financial transactions, climate data, etc.
- Big p . Large number of features relative to observations. (High-dimensional data).
 - ▶ Examples: Medical data - genomics, neuroimaging, medical imaging, etc.

Big Biomedical Data

Examples:

- High-throughput Genomics (“Omics”).
 - ▶ single-cell sequencing, RNA-sequencing, microarrays, methylation arrays, CGH-arrays, exome sequencing, mass spectrometry, NMR spectroscopy, etc.
- Neuroimaging / neural recordings.
 - ▶ MRI, Functional MRI (fMRI), EEG, MEG, DTI, ECoG, PET, calcium imaging, voltage imaging, electrophysiology, etc.
- Electronic Health Records.
- Medical Imaging.
- Text Data.

Data Matrix

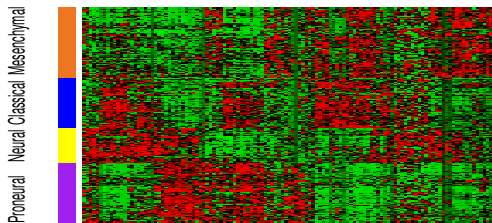
Data Matrix:

$$\mathbf{X}_{n \times p} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ \vdots & & \ddots & \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

- Rows: n observations / samples / subjects.
- Columns: p features / variables.

Data Matrix

Example: Omics Data



Gene Expression Data (Microarray)

- Rows (observations): Subjects ($n \approx 100 - 500$).
- Columns (features): Genes ($p \approx 500 - 20,000$).
- Measurement: Gene expression levels (loosely, how much a gene is turned off or on in a sample).

Data Matrix

Example: Text Mining

	data	R	big	cluster	shiny	fast	plot
doc 1	57	1	43	2	0	22	4
doc 2	17	29	2	3	35	6	44
doc 3	47	33	0	0	24	3	19
doc 4	23	0	0	31	0	7	2
doc 5	40	5	28	9	0	21	6
doc 6	8	10	7	46	12	17	9

(Bag-of-Words Format)

- Rows (observations): Documents ($n \approx 500 - 100,000$).
- Columns (features): Words ($n \approx 100 - 50,000$).
- Measurement: Count of how many times words appeared in documents.

Data Matrix

Example: Image Data



(Handwritten Digits Data)

- Rows (observations): Digits ($n \approx 10,000$).
- Columns (features): Pixels ($p = 256$).
 - ▶ Each digit image is converted to a 16×16 grayscale image. The 256 total pixels are vectorized to form the features.
- Measurement: Normalized grayscale intensity of each pixel.

Unsupervised vs. Supervised Learning

$$\mathbf{X}_{n \times p} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ \vdots & & \ddots & \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

- Rows: n observations / samples / subjects.
- Columns: p features / variables.

Supervised Learning:

$$\mathbf{y} = (y_1, y_2, \dots, y_n)^T$$

- $\mathbf{y}$ - n labels / outcomes associated with each observation.

Unsupervised Learning: No outcomes / labels!

Supervised Learning

Main Goal

Prediction!

- Given: $(Y_n^{train}, \mathbf{X}_{n \times p}^{train})$ (Training Data).
- Training: Use training data to find $\hat{f}()$ that maps $\mathbf{X}$ to Y :
 $Y = f(\mathbf{X}) + \epsilon$.
- Prediction: Given new $\mathbf{X}_{m \times p}^{test}$, predict $Y_{m \times 1}^{test}$: $\hat{Y}^{test} = \hat{f}(\mathbf{X}^{test})$.

Examples?

Secondary Goals:

- Feature Selection - What features are associated with the outcome?
- Others?

Unsupervised Learning

No labels! What is the goal?

Main Goal

Find some **structure** that characterizes the data.

(Or, find structure in training data that we expect to be present in future data.)

- Find patterns. (PCA, ICA, NMF, MDS)
- Dimension reduction. (PCA)
- Group observations / Group features / Group both. (Clustering)
- Find associations / relationships between features or observations. (Graphical or Network Models)
- Filter features. (Association testing)

Unsupervised Learning

Challenges:

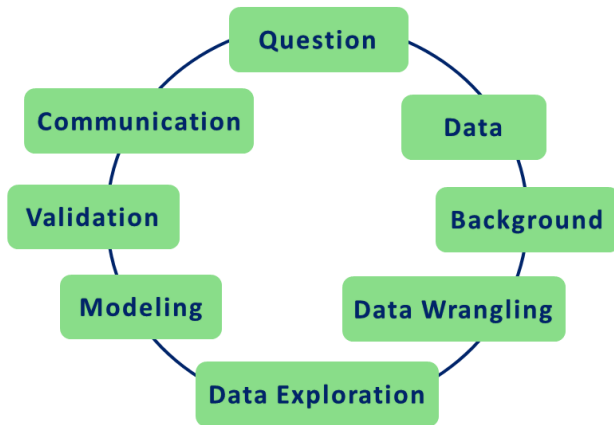
- Difficult to validate unsupervised learning results.
- No validation or test labels to measure prediction accuracy.
- What is meaningful structure in data?

Uses:

- Data pre-processing / compression / denoising.
- Exploratory data analysis.
 - ▶ Need to use multiple unsupervised learning techniques as each gives slightly different “insights” into data.
- Data visualization.
- Data-Driven Discovery.

Unsupervised Learning

How does it fit into a data science pipeline?



Unsupervised Learning

How is it used in Big Biomedical Data?

Case Study: BRCA gene expression data.

- Data Visualization.
 - ▶ Cluster heatmap, graphical models, MDS, PCA.
- Exploratory Analysis.
 - ▶ Clustering / dimension reduction to find cancer subtypes.
- Gene Selection.
 - ▶ Large-scale hypothesis testing to find genes associated with subtypes.
- Gene Interactions.
 - ▶ Graphical models.

Breakout Discussion

- How will you use Unsupervised Learning?
- What type of big data do you work with?
- What do you hope to learn from this course?

This Course

Day 1:

- ① Lecture 1: 8-8:50am - Intro
- ② Lecture 2: 9-9:50am - Dimension Reduction I
- ③ Lecture 3: 10-10:50am - Dimension Reduction II/Lab Intro
Break
- ④ Lecture 4: 11:30-12:20pm - Dimension Reduction III / Lab
- ⑤ Lecture 5: 12:30-1:20pm - Dimension Reduction Lab
- ⑥ Lecture 6: 1:30-2:20pm - Clustering I

**All times Pacific.*

This Course

Day 2:

- ① Lecture 1: 8-8:50am - Clustering II
- ② Lecture 2: 9-9:50am - Clustering III / Lab
- ③ Lecture 3: 10-10:50am - Clustering Lab
- Break
- ④ Lecture 4: 11:30-12:20pm - Graphical Models I
- ⑤ Lecture 5: 12:30-1:20pm - Graphical Models II
- ⑥ Lecture 6: 1:30-2:20pm - Testing

**All times Pacific.*

This Course

Day 3:

- ① Lecture 1: 8-8:50am - Validation & Workflow + Final Lab
- ② Lecture 2: 9-9:50am - Final Lab
- ③ Lecture 3: 10-10:50am - Final Lab Results + Best Practices

**All times Pacific.*